

Automated Multimodal Crop Disease Diagnosis

Using Remote Sensing Images

Beyond Visible Spectrum: AI for Agriculture 2026

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1. Introduction

Crop diseases are one of the main problems that affect agricultural production. Early disease detection can help farmers reduce crop loss and take action before the disease spreads. In this project, we worked on a deep learning task related to crop disease diagnosis using remote sensing data.

The project is based on the Beyond Visible Spectrum: AI for Agriculture 2026 Kaggle competition. The main goal is to classify crop health conditions using multimodal remote sensing images. The dataset contains three types of data: RGB images, multispectral images, and hyperspectral images. This makes the project more challenging than normal image classification because each modality gives different information about the crop.

The classification task includes three classes: Health, Other, and Rust. The main idea of our work was to understand the dataset, prepare the different modalities correctly, build baseline models, test fusion strategies, and compare the results in a clear way.

2. Problem Definition

The problem is to classify crop health conditions using deep learning models. The input data is not only normal RGB images. It also includes multispectral and hyperspectral images.

RGB images show visual information, but they may not capture hidden disease symptoms. Multispectral and hyperspectral images contain spectral information that can help detect plant stress or disease before it becomes clear to the human eye.

Problem statement: Given RGB, MS, and HS images of crop samples, build a deep learning model that classifies each sample as Health, Other, or Rust.

3. Objectives

1. Understand the structure of the multimodal dataset.
2. Preprocess RGB, MS, and HS images correctly.
3. Build two baseline deep learning models.
4. Compare the performance of the baseline models.
5. Build and test fusion models using more than one modality.
6. Evaluate all models using accuracy, precision, recall, F1-score, and confusion matrix.
7. Identify the best-performing model based on validation results.

4. Literature Review

Remote sensing has become an important tool in agriculture because it can collect crop information without touching the plants. In crop disease detection, normal RGB images can show visible symptoms, but some disease signs may appear first in spectral bands before they are clearly visible to the human eye.

Hyperspectral images are useful because they contain many narrow bands. This gives detailed information about the crop condition, but it also makes the data high-dimensional and harder to process. Multispectral images have fewer bands, but they are easier to handle and can still provide useful spectral information for classification.

Deep learning models, especially CNN models, are commonly used for image classification because they can learn visual and spatial features automatically. For spectral data such as MS and HS images, CNN models can also learn patterns between bands and help classify crop health conditions.

Previous work in crop disease detection shows that using more than one data source can be helpful, but fusion is not always simple. The different modalities may have different image sizes, different numbers of bands, and different information. Because of that, this project tested separate baseline models and fusion models to see which approach works better on this dataset.

5. Dataset Description

The dataset is organized into train and val folders. Each folder contains three modalities: HS, MS, and RGB.

Modality	File Type	Description
RGB	.png	Normal color images
MS	.tif	Multispectral images with 5 bands
HS	.tif	Hyperspectral images with many spectral bands

Table 1. Dataset modalities and file types.

The file count check showed that all three modalities have matching numbers of files.

Set	HS Files	MS Files	RGB Files
Train	600	600	600
Val	300	300	300

Table 2. File count by modality.

We also checked that the sample names match across HS, MS, and RGB. The result showed that all 600 training samples and all 300 validation samples are matched correctly across the three modalities.

6. Labels and Class Distribution

The labels were extracted from the training filenames. The training data contains three classes: Health, Other, and Rust.

Class	Number of Samples
Health	200
Other	200
Rust	200

Table 3. Original training class distribution.

The dataset was balanced in the training set. Each class had exactly 200 samples.

Since the original validation folder did not show labels clearly from the filenames, we created our own validation split from the training data using stratified splitting.

Class	Train	Validation
Health	160	40
Other	160	40
Rust	160	40

Table 4. Stratified train-validation split.

7. Preprocessing

7.1 RGB Preprocessing

RGB images were loaded using PIL. Each image was resized to 224 x 224 x 3. Pixel values were normalized by dividing by 255, so the values became in the range 0 to 1.

Data	Shape
Train RGB	(480, 224, 224, 3)
Validation RGB	(120, 224, 224, 3)

Table 5. RGB array shapes after preprocessing.

7.2 MS Preprocessing

MS images were stored as .tif files. Each MS image had the shape 64 x 64 x 5, which means each sample has 5 spectral bands. Since MS pixel values were not in the normal RGB range, min-max normalization was applied to each image.

Data	Shape
Train MS	(480, 64, 64, 5)
Validation MS	(120, 64, 64, 5)

Table 6. MS array shapes after preprocessing.

7.3 HS Preprocessing

HS images were also stored as .tif files. Most HS images had the shape 32 x 32 x 125. However, some HS images had 126 bands instead of 125. To keep the input shape consistent, all HS images were standardized to 125 bands by keeping the first 125 bands.

Data	Shape
Train HS	(480, 32, 32, 125)
Validation HS	(120, 32, 32, 125)

Table 7. HS array shapes after preprocessing.

8. Methodology

We implemented several models to compare the effect of different modalities. The models were simple enough to understand and compare, but each one was designed to test a clear idea.

- RGB Baseline Model
- MS Baseline Model
- MS + HS Early Fusion Model
- RGB + MS Late Fusion Model
- Improved MS Model

This approach helped us understand which modality gives better performance and whether fusion improves the results.

9. Baseline and Fusion Models

9.1 RGB Baseline

The first baseline used only RGB images. The model was a CNN with convolution layers, max pooling layers, batch normalization, global average pooling, dense layer, dropout, and softmax output.

Setting	Value
Input Shape	224 x 224 x 3
Optimizer	Adam
Loss	Sparse Categorical Crossentropy
Validation Accuracy	51.67%

Table 8. RGB baseline setup and result.

9.2 MS Baseline

The second baseline used only multispectral images. The input shape was 64 x 64 x 5. The model was similar to the RGB baseline, but it was adapted for 5-channel input.

Class	Precision	Recall	F1-score
Health	0.52	0.28	0.36
Other	0.75	0.75	0.75
Rust	0.54	0.80	0.65

Table 9. MS baseline classification report.

The MS baseline achieved 60.83% validation accuracy and was the best model in our experiments.

9.3 MS + HS Early Fusion

In this model, we tried to combine MS and HS data inside one network. The model had two branches: one for MS and one for HS. The extracted features were joined using concatenation.

Model	Validation Accuracy
MS + HS Early Fusion	30.83%

Table 10. Early fusion result.

This model performed poorly and predicted most samples as the Other class. This shows that simple concatenation was not enough for combining these modalities.

9.4 RGB + MS Late Fusion

Late fusion was tested by combining the prediction probabilities from the RGB and MS models. Different weights were tested between the two models.

Best RGB Weight	Best MS Weight	Best Accuracy
0.0	1.0	60.83%

Table 11. Late fusion best result.

The best late fusion result was the same as using MS alone. This means RGB did not improve the result in this experiment.

9.5 Improved MS Model

Since MS was the best modality, we tried an improved MS model using more convolution layers, batch normalization, L2 regularization, dropout, lower learning rate, and ReduceLROnPlateau.

Model	Validation Accuracy
Improved MS Model	27.50%

Table 12. Improved MS result.

The improved model performed worse than the simpler MS baseline. This may be because the model became too complex for the small dataset size.

10. Results Summary

Model	Validation Accuracy
RGB Baseline	51.67%
MS Baseline	60.83%
MS + HS Early Fusion	30.83%
RGB + MS Late Fusion	60.83%
Improved MS Model	27.50%

Table 13. Final model comparison.

The best-performing model was the MS Baseline model with 60.83% validation accuracy.

11. Discussion

The results show that MS data performed better than RGB data. This is expected because multispectral images contain spectral information that is not available in normal RGB images.

The RGB model struggled, especially with the Health class. This suggests that visual color information alone may not be enough for this disease classification task.

The early fusion model did not improve the performance. This shows that multimodal fusion is not always simple. Even though HS and MS contain useful information, combining them requires a better fusion strategy.

The improved MS model also did not improve the result. It was more complex, but the dataset size was small. This may have caused unstable learning or overfitting. Overall, the MS baseline was the most stable and best-performing model in our experiments.

12. Limitations

- The labeled training dataset is small, with only 600 samples.
- The original validation set did not have clear labels, so we created our own validation split.
- The fusion model was simple and used basic concatenation.
- We did not use advanced pretrained models because the data has non-standard input channels.
- The HS data had inconsistent band numbers, so preprocessing was needed to standardize it.

13. Conclusion and Future Work

In this project, we built and evaluated several deep learning models for crop disease classification using multimodal remote sensing data.

We tested RGB, MS, and fusion-based models. The best result was achieved by the MS baseline model with a validation accuracy of 60.83%.

The experiments showed that multispectral data was more useful than RGB data for this task. Fusion models did not improve the results in our current setup, which shows that multimodal fusion needs a more careful design.

Future Work

- Use 5-fold stratified cross-validation.
- Try pretrained backbones adapted for spectral data.

- Apply attention-based fusion between MS and HS.
- Use dimensionality reduction for HS bands.
- Try focal loss or other loss functions if imbalance appears in the test data.
- Improve the fusion model with feature alignment before concatenation.

14. References

1. Kaggle Competition: Beyond Visible Spectrum: AI for Agriculture 2026.
2. Han, L. et al. ICPR Competition on Beyond Visible Spectrum: AI for Agriculture.
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4. Chen, T. et al. A Simple Framework for Contrastive Learning of Visual Representations.
5. Hu, J. et al. Squeeze-and-Excitation Networks.
6. Liu, Z. et al. A ConvNet for the 2020s.
7. Zhang, X. et al. Deep learning-based yellow rust disease detection from hyperspectral UAV images.
8. Lin, T.-Y. et al. Focal Loss for Dense Object Detection.

Appendix: Work Completed

- Checked dataset structure.
- Counted files in HS, MS, and RGB folders.
- Checked file extensions.
- Checked matching files between HS, MS, and RGB.
- Checked sample names and labels.
- Created train DataFrame.
- Created stratified train-validation split.
- Loaded and preprocessed RGB images.
- Loaded and preprocessed MS images.
- Loaded and preprocessed HS images.
- Built and evaluated RGB baseline.
- Built and evaluated MS baseline.
- Built and evaluated MS + HS early fusion model.
- Tested RGB + MS late fusion.
- Tested improved MS model.
- Compared all results using accuracy, precision, recall, F1-score, and confusion matrices.

Appendix: Confusion Matrices

The following figure shows the confusion matrices for the main models tested in the project. These visualizations help explain which classes were correctly classified and where the models made mistakes.

Confusion Matrices for Validation Results

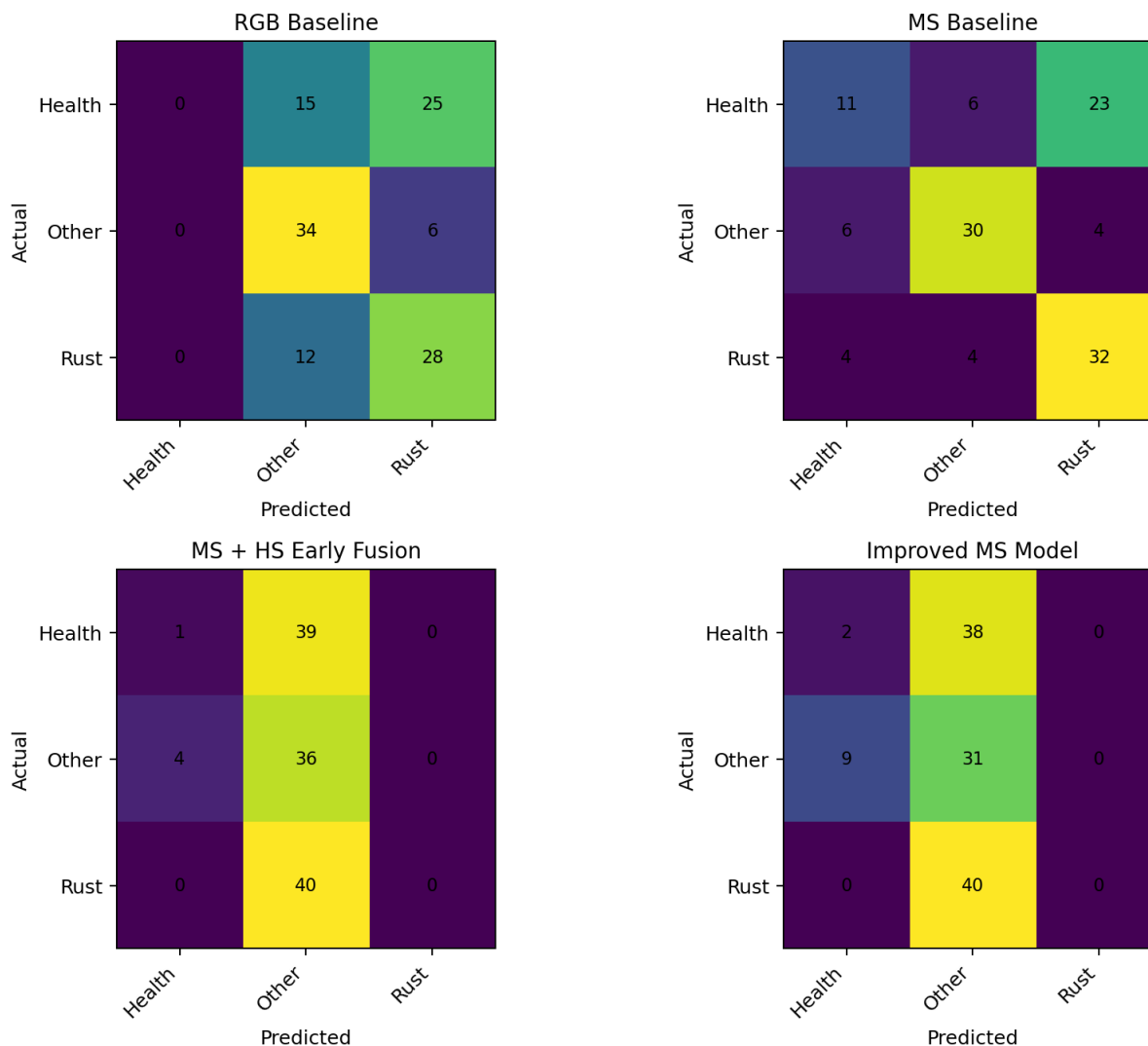


Figure 1. Confusion matrices for RGB baseline, MS baseline, MS + HS early fusion, and improved MS model.

Note. The MS baseline had the strongest result among the tested models, while the early fusion and improved MS models tended to misclassify many samples as the Other class.

Appendix: Kaggle Submission Result

✕ Submission Details



submission.csv

Complete (after deadline) · 3m ago

Score: 0.55789

Private score: 0.61463

UPLOADED FILES



submission.csv (6 KiB)



DESCRIPTION

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